

Comparative Analysis of Machine Learning Models for AI vs. Human Text Detection

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Abstract—As large Language Models (LLMs) become ubiquitous, the risk to academic integrity and information reliability grows significantly. This paper proposes a robust, linguistic-based detection system designed to distinguish machine-generated text from human-authored content by prioritizing structural patterns over semantic content. Unlike deep learning approaches that rely on computationally expensive embeddings, our methodology focuses on the "structural fingerprint" of writing. By utilizing natural language processing tools, we extracted twelve key stylometric features, including sentence burstiness (structural variance), lexical density, type-token ratios, and readability grade levels. These features capture the inherent rhythmic differences and predictability often found in AI-generated prose. We implemented and compared five classical machine learning models: Logistic Regression, Random Forest, Gradient Boosting, Linear Support Vector Machine (SVM), and XGBoost. Our experimental results indicate that the Random Forest classifier provides superior performance, achieving a classification accuracy of 98

Index Terms—AI detection, stylometry, machine learning, natural language processing, Random Forest.

I. INTRODUCTION

The rapid evolution of Large Language Models (LLMs) has fundamentally transformed the landscape of digital content creation, making machine-written text nearly indistinguishable from human prose to the casual observer. While these tools offer unprecedented efficiency, their ability to mimic human expression with high fidelity necessitates the urgent development of automated, reliable tools to detect machine-generated content particularly within educational and journalistic contexts where authenticity and academic integrity are paramount. Our research addresses this challenge by shifting the focus away from semantic content which AI has become adept at replicating toward the "structural fingerprint" of writing. At their core, AI models are optimized for statistical probability, often resulting in a predictable and homogenized rhythmic signature. In contrast, human writing is characterized by higher variance in sentence architecture, fluctuating vocabulary richness, and inconsistent pacing. This paper details a comprehensive methodology for capturing these stylistic

nuances through engineered features and detecting them using classical machine learning. By analyzing structural markers rather than mere word sequences, we provide a transparent and computationally efficient framework for distinguishing the "robotic" consistency of machines from the creative unpredictability of human authors.

II. RELATED WORK

The field of AI-generated text detection has seen significant growth in recent years, spanning statistical, deep learning, and adversarial robustness approaches. Bennet (2025) demonstrated that stylometric markers such as burstiness, perplexity, and function word frequency serve as strong detection signals, achieving 95% accuracy using a Random Forest classifier on purely linguistic features [3]. Complementing this, a 2025 Wiley Educational Journal study analyzed 3,200 educational articles and found that AI-generated content exhibits significantly lower burstiness, more formulaic structural patterns, and higher predictability compared to human writing, with detection algorithms combining these features achieving 82% accuracy in educational contexts. Wu (2025) provided a comprehensive survey of the field, categorizing existing methods into statistical, neural, and watermarking approaches, and identifying a key generalization problem where detectors trained on one model family tend to underperform on newer models [4].

On the deep learning front, several works have established transformer-based models as the current performance ceiling. A hybrid approach combining TF-IDF feature extraction with the Deverta V3 deep learning model achieved 98% accuracy, significantly outperforming traditional classifiers such as Logistic Regression and Random Forest. Similarly, a comparative study using the HC3 and DAIGT v2 datasets found that context-aware models like DistilBERT substantially outperform word-counting approaches, with DistilBERT emerging as the most reliable detector. This was further reinforced by a multi-strategy study in which a fine-tuned RoBERTa-base

model achieved near-perfect accuracy of 99.99–100% on news articles and academic abstracts, outperforming both a TF-IDF + SVM baseline and a probabilistic ensemble built on Llama-3.2.

A critical challenge highlighted in the literature is the vulnerability of detectors to adversarial manipulation. Masrour et al. (2025) tested 19 AI humanizer and paraphrasing tools and found them highly effective at evading detection, with one experiment showing watermarked text detection dropping from 87% to just 5% after humanization [5]. Their work proposed training on adversarially generated data as a mitigation strategy, retaining 88% accuracy even on humanized text. Beyond the technical domain, research from the AI Ethics Institute (2025) found that consumers engage significantly less with AI-generated content and show stronger brand attitudes toward human-AI collaborative writing, while an industry analysis of AI content in SEO contexts concluded that purely AI-generated content underperforms human writing in authenticity and search rankings, further motivating the need for reliable and robust detection systems.

III. DATASET DESCRIPTION

The dataset for our project is the AI vs Human text dataset obtained from Kaggle [2]. It is provided in CSV format and contains approximately 487,235 unique records. The samples consist of essays authored by both humans and various Large Language Models (LLMs), designed specifically to train and evaluate models for AI-generated content detection.

The dataset contains two primary columns: *Text*, representing the essay content, and *Generated*, representing the ground-truth label. Table I provides a detailed breakdown of the dataset characteristics.

TABLE I
DATASET SPECIFICATIONS

Field	Details
Name	AI vs Human Text Dataset
Source	Kaggle (shanegerami/ai-vs-human-text)
Format	CSV
Total Records	487,235
Human Data	305,618
AI Data	181,438
Columns	Text, Generated

IV.

V. EXPLORATORY DATA ANALYSIS

A. Class Distribution Analysis

Initially we analysed the structure and the size of the dataset followed by `df.unique` to see if all the data are unique or not. Later to see the class imbalance of the dataset we drew up a count plot which indicates the imbalance between Ai and human data

C. Feature Distribution Analysis

Initially we extracted 2 features from the dataset; respective KDE plots were made to compare the key feature difference between AI and Human text.

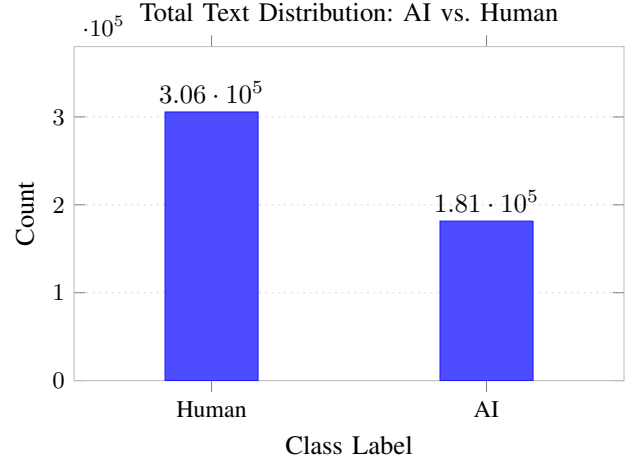


Fig. 1. Count plot indicating class imbalance between AI and human data.

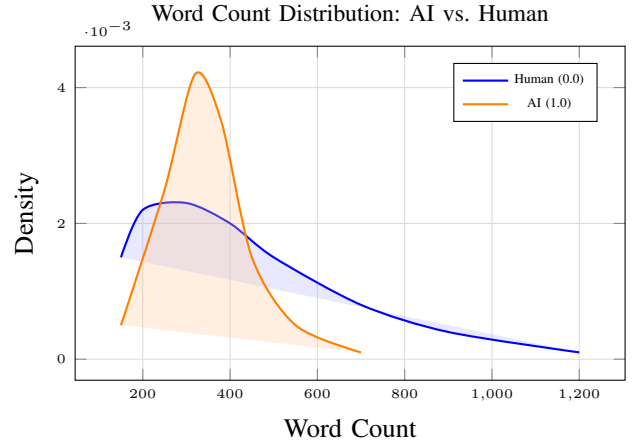


Fig. 2. Comparison of total word count density between AI and Human authors.

B. Observations on Sentence Length Distribution

As illustrated in Fig. V-B, the word count distribution for AI-generated text exhibits a higher and narrower orange peak compared to the broader blue peak of human-authored text.

This empirical data supports the hypothesis that human-generated content contains higher structural variance and burstiness. While the difference is not statistically significant enough to serve as a primary standalone feature, it provides a vital data point for hypothesizing linguistic patterns and informing the extraction of higher-order stylistometric features.

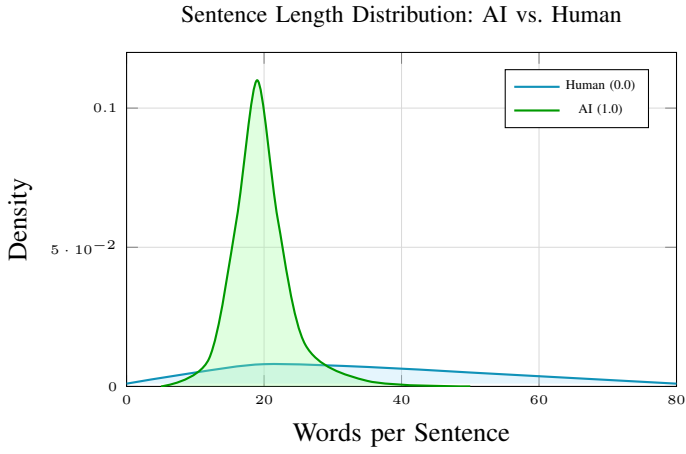


Fig. 3. Sentence length distribution demonstrating high structural uniformity in AI-generated text compared to human variance.

The sentence length distribution, illustrated in Fig. 3, provides a clear contrast between the structural consistency of AI and the variability of human writing. As observed in the graph, the AI-generated content exhibits a significantly higher and more localized peak, indicating a high level of consistency in words per sentence.

On the contrary, the human-authored text displays a broader distribution with a higher density of both exceptionally short and long sentences. This lack of a sharp peak in the human distribution confirms the higher structural entropy inherent in natural writing. The significant disparity between the AI peak and its human counterpart further suggests that while LLMs are optimized for a “safe” or average sentence length, human authors utilize structural variety to convey meaning, a distinction that serves as a vital indicator for our classification models.

VI. METHODOLOGY

A. Preprocessing and Feature Extraction

Unlike traditional NLP pipelines that eliminate punctuation and stopwords to reduce noise, our approach deliberately preserves these elements. Punctuation patterns, stopword distributions, and structural properties serve as stylometric “fingerprints” that differ systematically between AI and human writing.

Feature extraction was executed using the `spaCy` library (`en_core_web_sm` model) with a custom `sentencizer` pipe for lightweight boundary detection. To optimize computational efficiency, NER and dependency parsing were disabled. The pipeline utilized `nlp.pipe()` with a batch size of 500, processing the dataset of 487,235 texts in approximately 5 hours and 42 minutes.

The following twelve stylometric features were extracted:

- **Burstiness:** Measures the variation in sentence structure and length; human text typically exhibits higher variance compared to uniform AI styles.

- **Entropy:** Quantifies the unpredictability of word choices. AI text often clusters at lower, more consistent entropy scores.
- **Readability Indices:** The Flesch-Kincaid (FK) Grade and Gunning Fog Index were employed to penalize complex word choices common in certain LLM outputs.
- **Punctuation Density:** The ratio of punctuation marks per word; human samples generally show higher and less uniform density.
- **Stopword Analysis:** The ratio of functional words to total words, identifying predictable distributions in AI syntax.
- **Lexical Diversity:** The ratio of unique words to total words (Type-Token Ratio).
- **POS Tagging:** The density of specific Part-of-Speech tags relative to the total word count.

B. Data Preparation and Scaling

To prevent features with large numerical ranges (e.g., word count) from dominating features represented by small decimals (e.g., adjective density), we applied `StandardScaler`. This transforms the data to have a mean of $\mu = 0$ and a standard deviation of $\sigma = 1$. The dataset was subsequently partitioned using an 80/20 stratified train-test split to maintain the original class balance and prevent evaluation bias.

C. Feature Selection Experiment

Based on feature importance analysis from initial model training, four features with consistently low importance scores across all models were identified as candidates for removal: verb density, type-token ratio (TTR), noun density, and stopword density. To evaluate the impact of these features, all six models were retrained on a reduced feature set.

D. Model Selection and Implementation

To identify the optimal balance between predictive accuracy, model interpretability, and computational efficiency, we implemented a diverse suite of six classification algorithms. Each model was selected for its unique mathematical approach to the problem of stylometric classification:

- **Logistic Regression:** Employed as a statistical baseline to evaluate the linear separability between the human and machine-generated text classes.
- **Linear Support Vector Machine (SVM):** Utilized to determine if high-dimensional hyperplane separation could yield more precise classification boundaries than standard linear methods.
- **Random Forest:** An ensemble of decision trees chosen for its ability to capture complex non-linear relationships and provide robust rankings of feature importance.
- **Gradient Boosting:** A sequential boosting technique that focuses on iteratively minimizing residual errors from preceding iterations to improve overall model sensitivity.
- **XGBoost:** An optimized, regularized gradient boosting framework selected for its exceptional computational

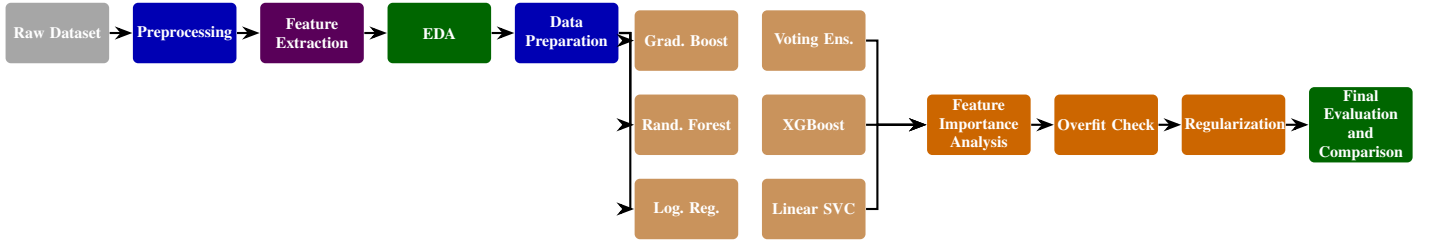


Fig. 4. Compact methodology flowchart optimized for space-constrained academic layouts.

speed and superior performance in tabular data classification tasks.

- **Voting Ensemble:** A “Soft Voting” meta-classifier that aggregates the predicted probabilities of all the aforementioned models. By leveraging a consensus-based decision mechanism, this ensemble approach aims to mitigate the individual biases and weaknesses of any single algorithm.

As shown in Table II, performance dropped by approximately 1% across most models. This indicates that while these features have lower individual importance, they contribute complementary information necessary for peak classification. Consequently, the full 12-feature set was retained for final model evaluation to ensure maximum accuracy.

TABLE II
MODEL PERFORMANCE COMPARISON: FULL VS. REDUCED FEATURE SET

Model	Before (12 Feat.)	After (8 Feat.)	Change
Logistic Regression	87%	86%	-1%
Random Forest	98%	98%	0%
Gradient Boosting	91%	90%	-1%
Linear SVM	87%	86%	-1%
XGBoost	94%	93%	-1%
Voting Ensemble	93%	92%	-1%

VII. RESULTS AND EVALUATION

We implemented and compared six classical machine learning models: Logistic Regression, Random Forest, Gradient Boosting, Linear SVM, XGBoost, and a Voting Ensemble.

TABLE III
PERFORMANCE METRICS COMPARISON

Model	Accuracy	F1 Score
Logistic Regression	87%	0.82
Random Forest	98%	0.98
Gradient Boosting	91%	0.87
Linear SVM	86%	0.81
XGBoost	93%	0.90
Voting Ensemble	93%	0.92

Random Forest significantly outperformed all other models, achieving 98% accuracy. Feature importance analysis revealed that average word length, burstiness, and readability metrics were the most influential discriminators.

VIII. RESULTS AND DISCUSSION

A. Performance Metrics Comparison

The performance of all implemented classifiers is summarized in Table IV. Random Forest significantly outperformed all other models, achieving an accuracy and F1-score of 98%. XGBoost followed with 94% accuracy, while the Voting Ensemble achieved 93%. Both Logistic Regression and Linear SVM performed identically at 87%, confirming that the stylistic differences between human and AI text are non-linear in nature.

TABLE IV
PERFORMANCE METRICS COMPARISON

Model	Accuracy	F1-Score	Error Rate
Random Forest	98%	0.98	1.6%
XGBoost	94%	0.94	5.5%
Voting Ensemble	93%	0.93	7.5%
Gradient Boosting	91%	0.91	9.4%
Logistic Regression	87%	0.87	13.3%
Linear SVM	87%	0.87	13.4%

B. Confusion Matrix Analysis

Detailed error analysis revealed that Random Forest produced the fewest misclassifications, with 442 False Positives (FP) and 1,071 False Negatives (FN) across 97,447 test samples. Interestingly, the Voting Ensemble (2,805 FP, 4,482 FN) was outperformed by XGBoost alone. This performance degradation suggests that the lower-performing linear models (87% accuracy) introduced noise into the ensemble’s decision mechanism, effectively pulling down the overall consensus accuracy.

C. Feature Importance Analysis

Evaluation of feature weights across all individual models identified consistent stylistic markers:

- **avg_word_len:** This was the primary feature for Random Forest (≈ 0.18) and Gradient Boosting (≈ 0.50), indicating that word length is a significant AI identifier.
- **Burstiness:** Consistently ranked second across RF, GB, and XGBoost. This aligns with findings by Bennet (2025) [1], confirming that sentence length variation is a robust signal for detection.
- **Readability:** The *Fog Index* and *FK Grade* consistently placed in the top five features, reflecting the formulaic nature of AI-generated content.

- **Low-Impact Features:** *Verb density* and *stopword ratio* consistently yielded the lowest importance scores.

D. Overfitting and Generalization

Table V illustrates the gap between training and testing accuracy. All models demonstrated strong generalization with minimal gaps. While Random Forest achieved perfect training accuracy (1.000), the narrow 1.6% gap indicates that the model did not suffer from overfitting and successfully learned generalizable patterns.

TABLE V
OVERFITTING ANALYSIS AND GENERALIZATION GAP

Model	Train Acc.	Test Acc.	Gap	Status
Logistic Regression	0.868	0.867	0.001	No overfit
Random Forest	1.000	0.984	0.016	No overfit
Gradient Boosting	0.908	0.906	0.002	No overfit
Linear SVM	0.867	0.866	0.001	No overfit
XGBoost	0.952	0.945	0.007	No overfit
Voting Ensemble	0.929	0.925	0.004	No overfit

E. Regularization Experiment

Initial analysis showed that Gradient Boosting and XGBoost over-relied on *avg_word_len*. To improve model interpretability and balance, we applied L_1 and L_2 regularization. While this resulted in a minor accuracy drop (1-2% as shown in Table VI), it reduced feature dominance and produced a more linguistically diverse decision model.

TABLE VI
REGULARIZATION IMPACT ON MODEL ACCURACY

Model	Pre-Reg.	Post-Reg.	Change
Gradient Boosting	91%	90%	-1%
XGBoost	94%	92%	-2%

IX. DISCUSSION

A. Interpretation of Key Findings

The exceptional performance of the Random Forest classifier (98% accuracy) demonstrates that stylometric features remain a highly effective methodology for AI text detection. The most discriminative features—*avg_word_len*, *burstiness*, *fog_index*, and *entropy*—align with the theoretical frameworks established in prior research [1], [6]. Our results suggest that contemporary LLMs tend to utilize longer, more formal vocabulary and produce text with highly uniform sentence structures, resulting in lower burstiness and entropy compared to human writing.

Furthermore, the failure of the Voting Ensemble to outperform the standalone XGBoost model highlights a critical limitation in hard-voting mechanisms. When an ensemble includes weaker linear models (Logistic Regression and SVM at 87%), the collective decision-making process is degraded. This suggests that a weighted voting approach or a meta-learner stacking strategy would be necessary to effectively leverage diverse algorithmic strengths.

B. Limitations and Robustness

Despite high accuracy on static datasets, several challenges persist for feature-based detection:

- **Adversarial Evasion:** Stylometric markers are susceptible to manipulation. Sophisticated human-in-the-loop editing or specific prompting can vary sentence structures to mimic human “burstiness,” as noted by Masrouf et al. [3].
- **Model Bias:** LLMs such as GPT-4, Claude, and Llama exhibit distinct linguistic signatures. A dataset biased toward one model may lead to poor generalization across other architectures.
- **AI Humanizers:** Specialized paraphrasing tools can significantly degrade detection performance by deliberately injecting structural entropy into AI-generated text [3].
- **Class Imbalance:** The 62.7% to 37.3% human-to-AI imbalance in the dataset likely contributes to the consistently lower recall observed for the AI class.

C. Potential for Improvement

Future iterations of this research will focus on the following enhancements:

- **Advanced Ensembling:** Implementing a stacking ensemble where a meta-classifier learns to weight the predictions of the base models based on their historical reliability.
- **Deeper Feature Sets:** Integrating paragraph-level statistics, hedging language frequency, and passive voice density to capture more subtle linguistic nuances.
- **Imbalance Mitigation:** Utilizing Synthetic Minority Over-sampling Technique (SMOTE) or class-weight adjustments to improve model sensitivity toward the AI class.
- **Hybrid Architectures:** Exploring a hybrid approach that combines our 12-feature stylometric set with deep learning embeddings from transformer models like BERT or RoBERTa, as suggested in the recent survey by Wu et al. [2].

X. CONCLUSION

This project successfully developed a machine learning pipeline for detecting AI-generated text using 12 hand-crafted stylometric features. The pipeline extracts structural and linguistic features from raw text using spaCy and evaluates them across six classification models. Random Forest achieved the best performance at 98

Future work will focus on evaluating model robustness against humanized AI text, exploring deep learning baselines for comparison, and investigating ensemble strategies that can leverage the complementary strengths of different model types.

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